

Not All Rollouts Teach: Contrast and Consensus in Online RL for Video World Models

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Abstract

Video world models should predict how a scene evolves under a specified physical event, yet online post-training on their own generated rollouts often yields limited gains. We identify two missing ingredients: rollouts must provide informative alternatives for comparison, and the learning objectives must agree on the resulting update. We call these requirements *contrast* and *consensus*. We introduce COSMIC (COntrastive Sampling and Multi-objectIve Consensus), an online reinforcement learning framework that operationalizes both principles. We first construct *PhysRL*, 1,371 target-free image–instruction conditions that pair a visible initial state with a specified physical event. For each condition, COSMIC generates structured alternative futures using antithetic initial-noise pairs, evaluates transition correctness and visual fidelity separately, and applies Sign Conflict-Aware NFT Optimization (SCAN) to suppress samples with conflicting reward-specific advantages. At the behavior-policy anchor, we show that each retained SCAN contribution is locally compatible with both reward objectives, whereas a sign-conflicting sample admits no nonzero compatible update along its NFT direction. With only 1.35K training conditions, COSMIC improves Cosmos-Predict2.5-2B from 81.0 to 82.0 in Overall Score and from 83.9 to 86.0 in Domain Score on PAI-Bench-G. It improves all six domains while maintaining comparable visual quality, and outperforms the official 14B post-trained model by 1.0 Domain points. Our results show that self-generated futures become effective supervision when they are organized to provide both contrast among rollouts and consensus among objectives. Code, models, and datasets will be released.

1 Introduction

“Uniformity does not lead to continuation; it is harmony that begets new things.”

— Shibo, Guoyu

Video generation models are increasingly used as *video world models*: given an observed scene and a transition instruction, they generate a short-horizon future trajectory (NVIDIA et al. 2025b; Team Wan et al. 2025; Google DeepMind 2025b; Team Seedance et al. 2026). This shift changes what counts as success. A plausible continuation may suffice for video generation; a world model must predict a future faithful to *this* scene and *this* requested event. It must preserve the observed state, realize the transition with plausible dynamics, and remain visually coherent. This capability

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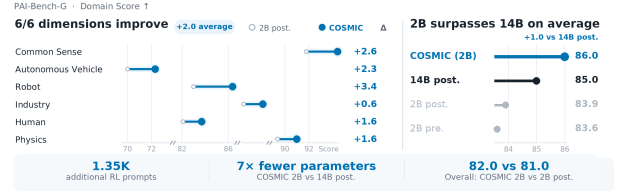


Figure 1: **Cross-domain transfer of COSMIC.** With only 1.35K conditions, COSMIC improves all six PAI-Bench-G domains at comparable Quality and exceeds the official 14B post-trained model.

supports policy learning (Yang et al. 2024; Jang et al. 2025; Jiang et al. 2026; Guo et al. 2026a), closed-loop driving simulation (Hu et al. 2023; Wang et al. 2024; Wu et al. 2025b; Lu et al. 2025), and interactive control (Google DeepMind 2025a; Ye et al. 2026; Tencent Hunyuan 2025; Kim et al. 2026; Zhang et al. 2026). Yet current models exhibit physical errors (Bansal et al. 2025, 2026; Meng et al. 2025), instruction drift (Guo et al. 2026b; Zhou et al. 2026a), and temporal or visual degradation (Huang et al. 2024; Zheng et al. 2025).

Online reinforcement learning (RL) seems a natural solution: sample several futures per condition, score them, and reinforce those that perform better within the group. Yet post-training gains remain modest or uneven (NVIDIA et al. 2025b; Chen et al. 2026). Sampling from the current model keeps rollouts relevant, but the comparison can fail in two ways. First, a small i.i.d. group may contain near-duplicates or repeat the same failure, revealing little about which behavior is better. Second, one rollout may follow the instruction better but look worse. A weighted sum forces a single ranking, but that ranking can change with the weights. We call meaningful variation *contrast* and agreement between criteria *consensus*; a rollout group is informative only when it provides both.

We operationalize this principle in COSMIC (COntrastive Sampling and Multi-objectIve Consensus). First, we construct *PhysRL*, 1,371 target-free image–instruction conditions with visually grounded and temporally explicit transitions. Second, antithetic initial noises (Jia et al. 2026) create structured alternatives while preserving each roll-

out’s marginal distribution without increasing rollout cost. Third, we keep a task-conditioned transition reward and a VideoAlign-based quality reward (Liu et al. 2025b) separate. **Sign Conflict-Aware NFT Optimization (SCAN)** then reinforces a rollout when both scores are above their group baselines, suppresses it when both are below, and omits it when they disagree.

COSMIC improves the Cosmos-Predict2.5-2B post-trained baseline on PAI-Bench-G (Zhou et al. 2026a), raising Overall Score from 81.0 to 82.0 and Domain Score from 83.9 to 86.0. The gains span all six domains while Quality remains comparable, and the resulting 2B model exceeds the official 14B post-trained model by 1.0 Domain points. Ablations support both sides of our design: pairwise antithetic sampling produces more effective rollout groups than i.i.d. sampling, while factorized task and quality rewards together with **SCAN** provide a better balance than single-reward or prematurely scalarized alternatives. Taken together, these results suggest that the value of on-policy rollouts depends not only on how many futures are sampled, but also on whether they expose informative alternatives and yield a coherent learning signal.

Our contributions are threefold:

- We identify *contrast* and *consensus* as two complementary properties of rollout-group supervision for online video world-model post-training: the sampled futures should reveal meaningful alternatives, and the retained objectives should provide compatible learning signals.
- We introduce COSMIC, which instantiates these properties through well-specified, target-free *PhysRL* conditions, antithetic alternative-future sampling, factorized transition and quality rewards, and **SCAN**. We further provide a local analysis showing why sign-conflicting samples do not admit a common nonzero scalar update at the behavior-policy anchor.
- We demonstrate consistent cross-domain improvements on PAI-Bench-G: a compact 1.35K-condition COSMIC stage improves all six domains at comparable Quality and enables a 2B model to exceed the official 14B post-trained model by 1.0 Domain points.

2 Preliminaries

2.1 Video Generation Models as Flow Models

Video generation models can be formulated as conditional flow models (Liu, Gong, and Liu 2023; Lipman et al. 2023). Given a clean video latent $\mathbf{x}_0 \sim p_{\text{data}}$, Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and condition c , define

$$\mathbf{z}_t = (1 - t)\mathbf{x}_0 + t\epsilon, \quad \mathbf{v} = \mathbf{x}_0 - \epsilon. \quad (1)$$

The model is trained to predict the velocity $\mathbf{v}$:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{\mathbf{x}_0, \epsilon, t} \left[\|\mathbf{v}_\theta(\mathbf{z}_t, t, c) - \mathbf{v}\|_2^2 \right]. \quad (2)$$

At inference time, a deterministic ODE sampler (Chen et al. 2018), denoted by $G_\theta(\epsilon; c)$, starts from $\mathbf{z}_1 = \epsilon$ and integrates $\dot{\mathbf{z}}_t = -\mathbf{v}_\theta(\mathbf{z}_t, t, c)$ from $t = 1$ to $t = 0$. Thus, for fixed θ and c , the initial noise uniquely determines the generated rollout.

2.2 Antithetic Noise in Flow Models

Antithetic sampling replaces two independent initial noises with a paired draw $(\epsilon, -\epsilon)$:

$$\mathbf{x}_0^+ = G_\theta(\epsilon; c), \quad \mathbf{x}_0^- = G_\theta(-\epsilon; c). \quad (3)$$

Jia et al. (2026) show that the learned velocity field often satisfies an approximate affine antisymmetry,

$$\mathbf{v}_\theta(\mathbf{x}_t, t, c) + \mathbf{v}_\theta(-\mathbf{x}_t, t, c) \approx 2\mathbf{c}_t, \quad (4)$$

where $\mathbf{c}_t$ is the condition-dependent center of the flow at time t . Since each Euler update is linear in $\mathbf{v}_\theta$, this relation is propagated through the sampling trajectory, yielding

$$\mathbf{x}_0^- \approx -\mathbf{x}_0^+ + \text{const}(c). \quad (5)$$

The paired outputs are therefore negatively correlated. Meanwhile, $-\epsilon \stackrel{d}{=} \epsilon$, so each output retains the same marginal distribution as a standard model sample. Prior work studies antithetic noise in image generation; we evaluate it for video generation, which is much higher-dimensional.

2.3 DiffusionNFT with Multiple Rewards

DiffusionNFT (Zheng et al. 2026) optimizes a flow model on the forward process using a negative-aware flow-matching loss, decoupling model training from the reverse-process rollout. We adopt the multi-reward formulation of WorldCompass (Wang et al. 2026). For a rollout group $\{\mathbf{x}^{(i)}\}_{i=1}^G$ scored by J rewards $\{s_j^{(i)}\}_{j=1}^J$, let μ_j and σ_j be the within-group mean and standard deviation of reward j . The corresponding advantage is

$$a_j^{(i)} = \frac{s_j^{(i)} - \mu_j}{\sigma_j}. \quad (6)$$

The per-reward advantages are aggregated into an *optimality probability*

$$r^{(i)} = \frac{1}{2} + \frac{1}{2} \text{clip} \left(\frac{\sum_{j=1}^J \lambda_j a_j^{(i)}}{Z}, -1, 1 \right), \quad (7)$$

where $\sum_{j=1}^J \lambda_j = 1$ and Z is a normalization constant.

Let $\mathbf{v}_{\text{old}} := \mathbf{v}_{\theta_{\text{old}}}(\mathbf{z}_t^{(i)}, t, c)$ and $\mathbf{v}_\theta := \mathbf{v}_\theta(\mathbf{z}_t^{(i)}, t, c)$ denote the EMA and current-policy predictions, respectively, where $\tilde{\epsilon}^{(i)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\mathbf{z}_t^{(i)} = (1 - t)\mathbf{x}^{(i)} + t\tilde{\epsilon}^{(i)}$ and $\mathbf{v}^{(i)} = \mathbf{x}^{(i)} - \tilde{\epsilon}^{(i)}$. DiffusionNFT defines the implicit positive and negative velocities

$$\begin{aligned} \mathbf{v}_\theta^+ &= (1 - \beta)\mathbf{v}_{\text{old}} + \beta\mathbf{v}_\theta, \\ \mathbf{v}_\theta^- &= (1 + \beta)\mathbf{v}_{\text{old}} - \beta\mathbf{v}_\theta. \end{aligned} \quad (8)$$

Its training objective is

$$\begin{aligned} \mathcal{L}_{\text{NFT}} = \mathbb{E}_{t, i} \left[r^{(i)} \left\| \mathbf{v}_\theta^+ - \mathbf{v}^{(i)} \right\|_2^2 \right. \\ \left. + (1 - r^{(i)}) \left\| \mathbf{v}_\theta^- - \mathbf{v}^{(i)} \right\|_2^2 \right]. \end{aligned} \quad (9)$$

COSMIC: Contrastive Sampling and Multi-objective Consensus

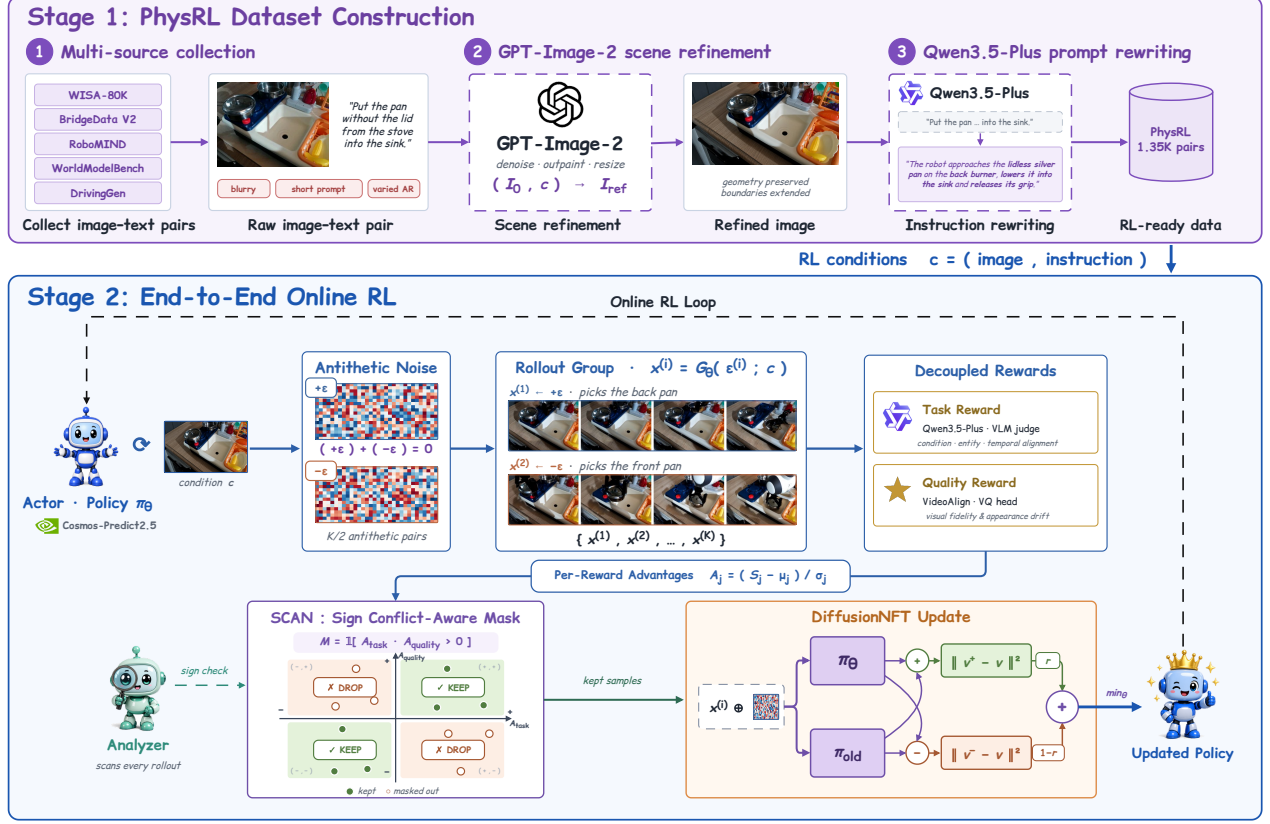


Figure 2: **Overview of COSMIC.** COSMIC makes finite rollout groups informative and actionable: *PhysRL* builds image–instruction conditions, antithetic noise creates contrasting rollouts, decoupled rewards preserve distinct objectives, and *SCAN* removes sign-conflicting samples before DiffusionNFT updates.

3 Method

3.1 *PhysRL* Dataset Construction

RL training for I2W (Image-to-World) generation requires only image–text pairs, without reference videos. However, each pair serves both as the condition for policy generation and as the context for reward-model evaluation. Existing sources exhibit three recurring problems. 1) A substantial fraction of the images are blurred or otherwise corrupted, obscuring task-relevant objects from the policy and reducing the reliability of VLM-based reward estimates. 2) Their prompts are often terse and underspecified, stating only the task objective while leaving the intended motion largely unconstrained. Consequently, the policy lacks sufficient guidance for physically plausible transitions, while the reward model lacks explicit criteria for evaluating them. 3) The pooled sources vary widely in resolution and aspect ratio, and naive resizing to a common format can deform salient structures such as robot arms and vehicles, corrupting the conditioning signal and destabilizing training.

To address these issues, we construct *PhysRL*, a compact, high-quality dataset for I2W RL (Stage 1 in Figure 2). It com-

prises 1,371 image–text pairs from WISA-80K (Wang et al. 2025), BridgeData V2 (Walke et al. 2023), RoboMIND (Wu et al. 2025a), WorldModelBench (Li et al. 2025), and DrivingGen (Zhou et al. 2026b). We filter the first three sources for annotation validity, camera stability, and visual quality, followed by perceptual-hash deduplication and quota-based diversity sampling; the latter two benchmarks are verified and adopted directly. We then use GPT-Image-2 (OpenAI 2026b) to mitigate visual degradation and outpaint each image onto a common 1280×704 canvas matching the I2W model’s native resolution. A fixed editing prompt requires the camera viewpoint, objects, and spatial layout to remain unchanged, preserving scene geometry without naive resizing. Finally, conditioned on each standardized image, Qwen3.5-Plus (Qwen Team 2026) rewrites the annotation under domain-specific templates into a single paragraph that grounds the visible initial state and describes how the intended action or physical process unfolds over time. The resulting conditions combine uniform geometry, high visual fidelity, and temporally explicit instructions, enabling more reliable reward evaluation and stable I2W policy optimization. Further details are provided in Appendix A.

3.2 COSMIC: COntRastive Sampling and Multi-objective Consensus

As illustrated in Figure 2, each online RL iteration samples a rollout group under condition c , evaluates the rollouts with decoupled task and quality rewards, and converts their group-relative advantages into a DiffusionNFT update. COSMIC makes this supervision informative and actionable: antithetic noise creates within-group contrast, and SCAN retains only samples whose two advantages agree in sign.

Antithetic Rollout Noise Group-relative optimization relies not only on the marginal quality of individual rollouts, but also on how the finite rollout group is organized. Under a shared image-text condition c , i.i.d. Gaussian sampling produces valid rollouts $G_\theta(\epsilon; c)$ (§2.1), but it does not explicitly encourage diversity. A small group may therefore contain redundant futures that expose similar behaviors or failure modes, providing limited contrast for relative evaluation.

We introduce structure into the rollout group through antithetic noise. For an even group size G , we replace the G i.i.d. draws with $G/2$ pairs of the form $(\epsilon, -\epsilon)$ introduced in Equation 3:

$$\{\epsilon^i\}_{i=1}^G = \{z_1, \dots, z_{G/2}, -z_1, \dots, -z_{G/2}\}, \quad (10)$$

$$z_j \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

Since $-z \stackrel{d}{=} z$, every $\epsilon^{(i)}$ remains marginally standard Gaussian. Antithetic sampling therefore preserves the distribution of each individual rollout while altering the *joint* structure of the finite group. This distinction is central to our design: the goal is not to change what constitutes a valid model sample, but to organize the samples under the same condition into more informative alternatives. The approximate affine antisymmetry in Equation 4 motivates this pairing. Sampling from z and $-z$ tends to drive the corresponding trajectories in opposing directions around a condition-dependent center.

This construction is compatible with the forward-process formulation of DiffusionNFT. The antithetic variables $\epsilon^{(i)}$ determine the generated rollouts included in the group, whereas each rollout is subsequently re-noised using an independent $\tilde{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ when constructing the training state for Equation 9. Thus, the antithetic dependence is introduced at rollout-group construction without replacing the independent re-noising used by the NFT objective. This argument is specific to forward-process optimization and does not apply to reverse-trajectory objectives such as Flow-GRPO (Liu et al. 2025a).

Decoupled Multi-Objective Rewards The reward model determines the direction of policy optimization, and a poorly specified reward can induce reward hacking. This risk is amplified in I2W generation because the objective is intrinsically *multi-objective*; optimizing a single axis such as instruction following almost invariably degrades visual fidelity. We therefore decouple the supervisory signal into two complementary components.

Given the condition image c , prompt p , and generated video $\mathbf{x}$, a VLM evaluator produces the scalar task reward

$$r_{\text{task}} = f_{\text{VLM}}(\mathcal{T}_{\text{task}}(c, \mathbf{x}, p)) \in [0, 1],$$

where $\mathcal{T}_{\text{task}}$ denotes the evaluation template. The evaluator jointly assesses *condition consistency*, *entity correctness*, and *action-temporal alignment*, measuring whether the video follows the instruction and faithfully completes the intended physical event. We use Qwen3.5-plus (Qwen Team 2026) as the task-reward because of its ability to extract fine-grained temporal relationships. The full template and further details are provided in Appendix C.

The task reward governs *what* the video depicts but provides no incentive for *how well* it is depicted. Because **PhysRL** provides condition images of relatively high visual quality, appearance drift is immediately perceptible; moreover, optimizing the task reward alone tends to produce physically plausible yet visually suboptimal solutions. We therefore use the visual-quality head of VideoAlign (Liu et al. 2025b) as the quality reward r_{quality} .

Sign Conflict-Aware NFT Optimization The task and quality rewards supervise distinct objectives and differ substantially in scale and dynamic range. Equation 6 therefore group-normalizes them independently into advantages $a_t^{(i)}$ and $a_q^{(i)}$, after which Equation 7 aggregates the two advantages into an optimality probability.

This composition exposes a pervasive yet under-examined failure mode. Empirically, a substantial fraction of samples within each group have *sign-conflicting* advantages across the two reward dimensions. Such samples are detrimental in two compounding ways: (i) the advantages partially cancel, pulling the optimality probability toward the neutral value of 0.5 and attenuating the effective gradient magnitude; and (ii) they provide conflicting supervision across the two reward dimensions, making them unreliable training instances for scalarized NFT optimization.

To resolve this conflict, we propose *Sign Conflict-Aware NFT Optimization (SCAN)*. SCAN examines every rollout for advantage-sign disagreement and masks conflicting samples before constructing the NFT training loss. Analyzing Equation 9 shows that the negative gradient contributed by rollout i factorizes into a reward-dependent scalar and a reward-independent vector:

$$-\nabla_{\theta} \mathcal{L}_{\text{NFT}}^{(i)} = b_i h_i \quad (11)$$

Here, $b_i = 2r^{(i)} - 1 \in [-1, 1]$ is the clipped scalarized advantage. The vector h_i is defined as

$$h_i := 2\beta J_i^\top (v^{(i)} - v_{\text{old}}) \quad (12)$$

where J_i denotes the model Jacobian $\frac{\partial v_\theta}{\partial \theta}$, $v^{(i)}$ is the target velocity, and v_{old} is the frozen EMA prediction. Crucially, h_i specifies a shared gradient direction determined entirely by the model parameters and re-noising draw, with no explicit dependence on reward signals or aggregation weights.

Because h_i is fixed for both objectives, any non-zero scalar coefficient b_i inevitably follows one reward at the expense of the other when their advantages disagree. SCAN therefore applies a binary mask m_i according to advantage-sign agreement before the gradient step:

$$m_i = \mathbb{1}[a_t^{(i)} a_q^{(i)} > 0] \quad (13)$$

Under the local NFT factorization, the resulting masked NFT signal is

$$\Delta_i^{SCAN} = m_i b_i h_i \quad (14)$$

This construction yields a simple mask-consistency property: every retained sample has reward advantages with compatible signs, and its scalarized NFT coefficient is aligned with each nonzero per-reward advantage. Let $g_{ij} = a_j^{(i)} h_i$ denote the objective-specific local preference vector for reward $j \in \{t, q\}$. For every retained rollout ($m_i = 1$), the retained NFT signal is aligned with both reward-specific local preferences:

$$\langle \Delta_i^{SCAN}, g_{ij} \rangle = m_i b_i a_j^{(i)} \|h_i\|_2^2 \geq 0, \quad \forall j \in \{t, q\} \quad (15)$$

Conversely, for any sign-conflicting sample ($a_t^{(i)} a_q^{(i)} < 0$) with $h_i \neq 0$, the zero update is the only update in the sample-induced scalar subspace that is a first-order non-ascent direction for both reward-specific local NFT surrogates. Under the binary keep-or-drop decision, masking is therefore the conservative max-min choice with respect to the worst-case reward-specific directional gain; it is unique whenever $b_i h_i \neq 0$. The formal gradient factorization and constrained optimality analysis are provided in Appendix E.

4 Experiments

4.1 Experiment Setup

Training Setup. We apply COSMIC to the official post-trained Cosmos-Predict2.5-2B checkpoint (NVIDIA et al. 2025b) using *PhysRL*. In our evaluations, each model generates 93-frame clips at 16 FPS. Implementation details are provided in Appendix D.

Evaluation Setup. We evaluate on PAI-Bench-G (Zhou et al. 2026a), developed from PBench (NVIDIA 2025) and used by successive Cosmos models. Its 5636 questions span common sense (CS), autonomous driving (AV), robotics (RO), industry (IN), human activity (HU), and physics (PH). We report the VQA-based *Domain Score*, evaluated with Qwen3-VL-235B (Bai et al. 2025) and GPT-5.4-mini (OpenAI 2026a) and corresponding *Quality Score*. Furthermore, we report the human-evaluated pairwise win-tie-loss rates with baselines.

Baselines. We compare with pre-trained and post-trained Cosmos-Predict2.5 at 2B and 14B, and with ABot-PhysWorld (Chen et al. 2026). For broader context, we include the official PAI-Bench-G results for Veo3, HunyuanVideo-I2V (Kong et al. 2024), CogVideoX1.5 (Yang et al. 2025), Cosmos-Predict2 (NVIDIA et al. 2025a), and the Wan family (Team Wan et al. 2025).

4.2 Main Results

Overall Performance. Table 1 shows that COSMIC raises the post-trained 2B backbone from 81.0 to 82.0 Overall and from 83.9 to 86.0 Domain. The gains span all six domains, led by robotics (83.0 $\rightarrow$ 86.4) and common sense (91.8 $\rightarrow$ 94.4), while Quality remains comparable (77.9 vs. 78.1). The resulting 2B model also exceeds the post-trained 14B checkpoint in

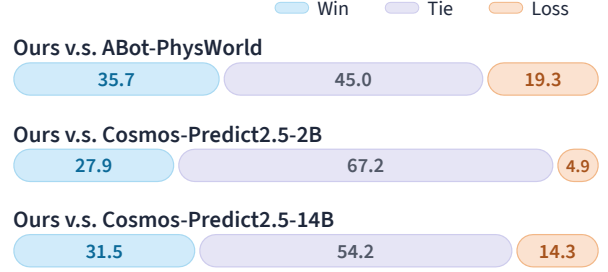


Figure 3: Pairwise preference against three baselines.

Overall and Domain scores (82.0/86.0 vs. 81.6/85.0), while approaching the strongest 27B and proprietary models.

The pairwise results in Figure 3 agree with the aggregate scores: COSMIC records more wins than losses against every baseline. Its win-or-tie rates against ABot-PhysWorld and the post-trained Cosmos-Predict2.5-2B and 14B models are 80.7%, 95.1%, and 85.7%, respectively.

Data Efficiency. As shown in Table 2, starting from the official post-trained 2B checkpoint, COSMIC uses only 1.35K additional RL conditions to gain 1.0 Overall and 2.1 Domain points. It thereby outperforms the 7 $\times$ -larger post-trained 14B model in both Overall and Domain. By contrast, ABot-PhysWorld uses 768K SFT and 64K RL samples but lowers its Wan2.1-I2V-14B backbone’s Overall and Domain scores by 0.5 and 0.9 points, respectively. Together, these results show that a compact, well-specified condition set can support effective online RL adaptation.

4.3 Qualitative Results

Figure 4 presents a representative comparison with ABot-PhysWorld and Cosmos-Predict2.5-2B/14B. ABot-PhysWorld produces an implausible interaction in which the robotic gripper crushes and deforms the package instead of grasping and lifting it. Both Cosmos-Predict2.5 variants also fail to complete the instructed action, revealing limitations in instruction following and physical transition prediction. In contrast, our model preserves the package structure and surrounding scene while executing the requested grasp-and-lift transition. This example illustrates its stronger alignment with both the specified event and the physical state of the scene. Additional qualitative comparisons are provided in Appendix F.

4.4 Ablation Study

All ablations use the post-trained Cosmos-Predict2.5-2B backbone with otherwise identical settings and isolate condition refinement, reward design, rollout-group construction, and multi-objective optimization.

Decoupled Rewards. Table 3 shows why both task correctness and visual quality are needed. With Qwen3.5+ as the task evaluator, adding the quality reward improves Overall, Domain, and Quality from 81.3/85.2/77.4 to 82.0/86.0/77.9. Quality-only optimization achieves the highest Quality score

Model	Params	Overall	Dimension Scores						Averages	
			CS	AV	RO	IN	HU	PH	Domain	Quality
Proprietary Models										
Veo3	–	82.2	93.0	72.1	88.8	89.0	84.4	89.7	86.8	77.6
Open-Source Models										
HunyuanVideo-I2V	13B	77.6	86.7	60.3	68.9	82.4	75.8	85.8	77.1	78.0
CogVideoX1.5	5B	78.4	88.7	64.4	73.6	84.6	79.1	88.7	80.3	76.6
Cosmos-Predict2	2B	79.7	91.2	70.7	81.9	86.4	83.1	88.1	84.2	75.2
Cosmos-Predict2	14B	80.0	92.5	69.8	81.3	86.6	82.4	89.2	84.3	75.8
Wan2.1-I2V	14B	80.8	92.5	69.4	79.6	87.2	83.4	89.5	84.3	77.3
Wan2.2-TI2V	5B	81.3	92.2	69.1	82.9	86.4	83.3	90.3	84.7	77.9
Wan2.2-I2V	27B	82.3	94.1	73.0	86.8	88.6	84.5	92.7	87.1	77.5
Our Evaluations										
Wan2.1-I2V	14B	80.6	93.7	66.1	82.9	87.0	82.4	90.9	83.8	77.3
Wan2.1-I2V + ABot-PhysWorld	14B	80.1	93.5	66.9	83.0	83.8	80.3	89.8	82.9	77.4
Cosmos-Predict2.5 [pre.]	14B	81.2	91.4	70.6	83.4	86.4	83.9	90.8	84.4	77.9
Cosmos-Predict2.5 [post.]	14B	81.6	92.8	69.8	85.4	87.8	83.5	90.9	85.0	78.2
Cosmos-Predict2.5 [pre.]	2B	80.7	91.2	71.2	81.7	86.5	82.2	89.0	83.6	77.8
Cosmos-Predict2.5 [post.]	2B	81.0	91.8	70.0	83.0	87.3	82.1	89.4	83.9	78.1
Cosmos-Predict2.5 [post.] + Ours	2B	82.0	94.4	72.3	86.4	87.9	83.7	91.0	86.0	77.9

Table 1: **Evaluation results on PAI-Bench-G.** Proprietary and open-source results are taken from the official benchmark; “Our Evaluations” reports our runs. Bold marks the best matched 2B Cosmos-Predict2.5 result per column.

Method	Training Samples			PAI-Bench-G		
	Pre.	SFT	RL	Overall	Domain	Quality
<i>Cosmos-Predict2.5-2B</i>						
[pre.]	200M	–	–	80.7	83.6	77.8
[post.]	–	17M	8K	81.0 (+0.3)	83.9 (+0.3)	78.1 (+0.3)
+ Ours	–	–	1.35K	82.0 (+1.0)	86.0 (+2.1)	77.9 (-0.2)
<i>Cosmos-Predict2.5-14B</i> (7× params)						
[pre.]	200M	–	–	81.2	84.4	77.9
[post.]	–	17M	8K	81.6 (+0.4)	85.0 (+0.6)	78.2 (+0.3)
<i>Wan2.1-I2V-14B-480P</i> (7× params)						
[pre.]	–	–	–	80.6	83.8	77.3
+ ABot-PW	–	768K	64K	80.1 (-0.5)	82.9 (-0.9)	77.4 (+0.1)

Table 2: **Incremental data efficiency.** Starting from the post-trained 2B checkpoint, our method achieves the best Overall and Domain scores among the compared adaptation settings with 1.35K additional RL conditions.

(78.1) but gives the lowest Domain score (84.5). Using Qwen3-VL + together with the quality reward matches the full configuration in Domain (86.0), but yields lower Overall and Quality scores (81.6/77.2 vs. 82.0/77.9). The full configuration therefore provides the best overall balance.

Rollout-Group Construction. Table 4 isolates how the finite rollout group is organized. In K -antithetic sampling, all K group members are jointly constructed with equal pairwise noise correlation $-1/(K-1)$ (Jia et al. 2026); our pair-

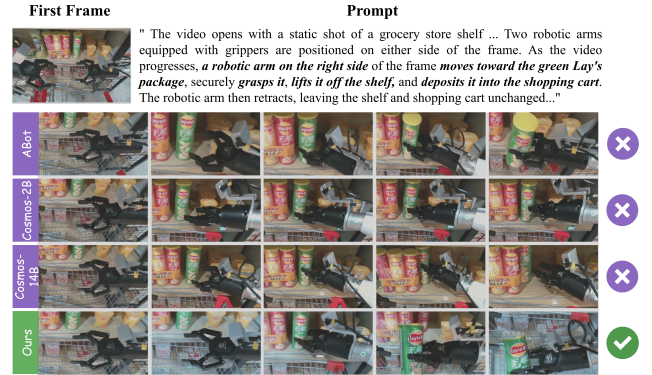


Figure 4: **Qualitative comparison.** The right robotic arm grasps the green Lay’s package and lifts it from the shelf.

wise variant instead uses independent $(z, -z)$ pairs. Pairwise sampling achieves the highest average Domain score (86.0), improving over i.i.d. sampling by 0.9 points and K -antithetic sampling by 0.4 points. Although no sampler dominates every domain, the pairwise construction gives the strongest aggregate result, supporting the practical value of structuring each finite rollout group with opposed pairs.

Condition Refinement and SCAN. In Table 5, *Original* denotes the conditions before GPT-Image-2 processing and Qwen3.5-Plus rewriting. With SCAN fixed, the final **PhysRL** conditions improve five of the six domains and raise the average Domain score from 85.0 to 86.0, despite a small decrease in physics (91.5 to 91.0). With **PhysRL** fixed, SCAN likewise improves five domains over EMAR (EMA Reward) and raises the average from 85.1 to 86.0, while physics decreases

Task RM	Quality RM	Overall	Domain	Quality
Qwen3-VL ⁺	✓	81.6	86.0	77.2
Qwen3.5 ⁺	–	81.3	85.2	77.4
–	✓	81.3	84.5	78.1
Qwen3.5 ⁺	✓	82.0	86.0	77.9

Table 3: **Reward ablation.** Qwen3-VL⁺ and Qwen3.5⁺ denote qwen3-vl-plus and qwen3.5-plus; VideoAlign (Liu et al. 2025b) provides the quality reward. The last row is our full configuration.

Noise	CS	AV	RO	IN	HU	PH	Avg
Pairwise	94.4	72.3	86.4	87.9	83.7	91.0	86.0
i.i.d.	94.8	71.0	84.7	86.3	83.4	90.1	85.1
K -antithetic	94.4	69.1	86.4	87.6	83.8	92.5	85.6

Table 4: **Rollout-noise ablation.** We compare pairwise antithetic (ours), i.i.d., and K -antithetic sampling (Jia et al. 2026).

from 91.4 to 91.0. These comparisons separately support condition refinement and reward-wise consensus.

5 Related Work

5.1 Video World Models

Video models (NVIDIA et al. 2025b; Team Wan et al. 2025; Google DeepMind 2025b; Team Seedance et al. 2026; Kong et al. 2024; Yang et al. 2025) have substantially advanced video synthesis and are increasingly deployed as world models for embodied learning (Yang et al. 2024; Jang et al. 2025; Jiang et al. 2026; Guo et al. 2026a; Huang et al. 2025a,b), closed-loop driving (Hu et al. 2023; Wang et al. 2024; Wu et al. 2025b; Lu et al. 2025), and interactive world modeling (Google DeepMind 2025a; Ye et al. 2026; Tencent Hunyuan 2025; Kim et al. 2026). Despite their visual appeal, reliably predicting a requested future remains challenging. Generated clips may violate object permanence, contact dynamics, or spatial relations, drift from the specified state change, or lose temporal and visual coherence. These failures are particularly acute in Image-to-World settings, where the predicted future must remain faithful to both the observed initial state and the requested physical event. Our work studies online post-training for this setting, seeking to improve transition correctness without sacrificing visual fidelity.

5.2 Reinforcement Learning for Video Generation Models

RL has driven recent breakthroughs in large language models (DeepSeek-AI et al. 2025; OpenAI 2025), and the same paradigm has been adapted to diffusion models (Black et al. 2024; Fan et al. 2023; Zheng et al. 2026; Liu et al. 2025a; Xue et al. 2025). Extending these techniques to video generation, however, is far from straightforward: expensive video rollouts and the lack of mature video reward models make naive online RL prohibitively costly. To avoid on-policy sam-

Conditions	Opt.	CS	AV	RO	IN	HU	PH	Avg
Original	SCAN	92.7	70.0	85.3	87.5	83.0	91.5	85.0
<i>PhysRL</i>	SCAN	94.4	72.3	86.4	87.9	83.7	91.0	86.0
<i>PhysRL</i>	EMAR	93.6	71.7	83.9	87.3	82.8	91.4	85.1

Table 5: **Condition and optimization ablations.** Adjacent-row comparisons isolate condition refinement and multi-objective optimization, respectively.

pling, recent work (Chen et al. 2026; Cai et al. 2026) curates offline preference data and applies DPO-style updates. However, supervision from a fixed preference distribution can become increasingly mismatched with the evolving policy, particularly in video, where preference pairs are scarce and expensive (Xu et al. 2025). NVIDIA et al. (2025b) explores online GRPO for video-world simulation but reports only modest gains over the pre-trained baseline. Beyond reducing rollout cost, effective online RL also depends on the supervision extracted from each finite rollout group. Our work frames this problem through *contrast* and *consensus*: sampled futures should expose informative alternatives, while the retained learning signal should be supported by both transition correctness and visual quality.

6 Conclusion

We presented COSMIC, an online RL framework for Image-to-World generation built around a simple premise: rollout-group supervision is effective only when the group exposes informative alternatives and the resulting update is supported by all optimization objectives. *PhysRL* provides clear, geometry-preserving conditions shared by rollout generation and reward evaluation. Upon this foundation, antithetic sampling introduces structured within-group contrast without altering the marginal rollout distribution, while factorized task and quality rewards expose objective disagreement and SCAN admits only sign-consistent updates. On PAI-Bench-G, COSMIC improves all six domains, raising the 2B post-trained model’s Domain Score from 83.9 to 86.0 while maintaining comparable Quality, and surpasses the official 14B model by 1.0 Domain points. These results suggest that effective online RL for world simulation depends not merely on sampling more futures, but on constructing useful comparisons and determining which comparisons should drive learning. As a general optimization framework, COSMIC is not tied to a particular dataset, and we hope to validate its effectiveness across a broader range of I2W datasets in future work. Due to computational constraints, our current evaluation focuses on the 2B backbone, while extending COSMIC to larger-scale models is left for future work.

Use of AI Tools

We use AI-assisted writing tools exclusively for improving language clarity and grammar. The scientific content, experiments, and conclusions were produced by the authors.

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